Class 19: Pertussis Mini Project

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Backround

Pertusis (a.k.a Whoopin cough) is a highly unfections lung infection caused by the bacteria *B. pertussis*

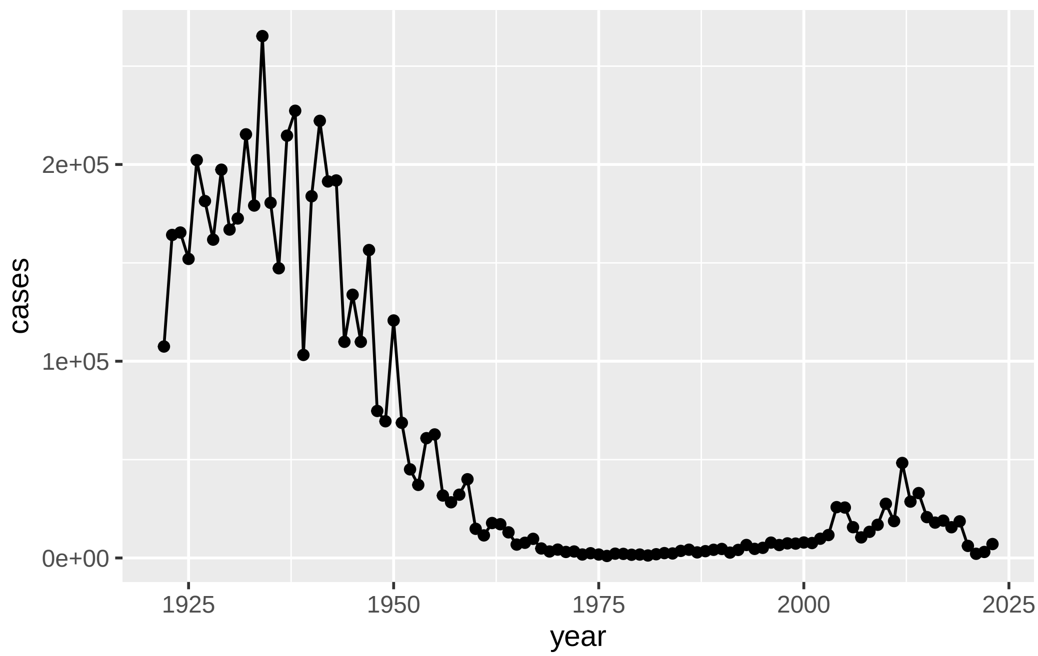
The CDC tracks case numbers in the US and makes this data available online:

> Q1. Make a plot of Pertussis cases per year with ggplot

```r
library(ggplot2)
```

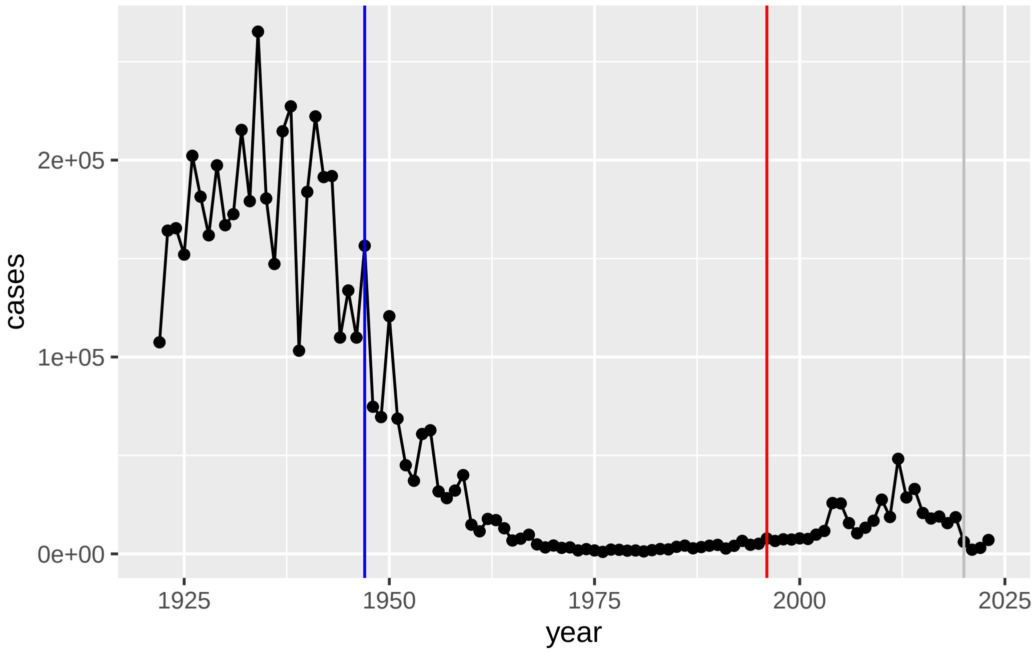
```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```r
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
```



Q2. Add some annotation (lines on the plot) for some major milestones in our interaction with Pertussis. The original wP deployment in 1947 and the newer aP vaccine roll-out in 1996. Finally a line for 2020.

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1947, col="blue") +
  geom_vline(xintercept = 1996, col="red") +
  geom_vline(xintercept = 2020, col="gray")
```



wP vaccine confers longer immunity than the aP vaccine, which needs a booster after 10 years, that's why after 1996 we can see a resurgence.

The CMI-PB project

The CMI-Pertussis Boost (PB) project focuses on gathering data on this very topic. What is distinct between aP and wP individuals over time when they encounter Pertussis again.

They make their data available via a JSON format returning API. We can read JSON format with the `read_json()` function from the **jsonlite** package.

```r
library(jsonlite)
subject <- read_json("http://cmi-pb.org/api/v5_1/subject",
                     simplifyVector = TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
```

```
5          5          wP          Male Not Hispanic or Latino Asian
6          6          wP        Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

> Q3. How many "subjects" (or individuals) are in this dataset?

```r
nrow(subject)
```

```
[1] 172
```

> Q4. How many wP and aP primmed subjects are there in the dataset?

```r
table(subject$infancy_vac)
```

```

aP wP 
87 85 
```

> Q5. What is the `biological_sex` and `race` breakdown of these subjects?

```r
table(subject$race, subject$biological_sex)
```

```
                                           
                                            Female Male
  American Indian/Alaska Native                  0    1
  Asian                                         32   12
  Black or African American                      2    3
  More Than One Race                            15    4
  Native Hawaiian or Other Pacific Islander      1    1
  Unknown or Not Reported                       14    7
  White                                         48   32
```

Let's read more tables from the CMI-PB database API

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)

ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

A wee peak at these:

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

Join (or link, or merge) using the `inner_join()` function from **dplyr**.

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
ab_data <- inner_join(meta, ab_titer)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(ab_data)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           1
3    1986-01-01    2016-09-12 2020_dataset           1
4    1986-01-01    2016-09-12 2020_dataset           1
5    1986-01-01    2016-09-12 2020_dataset           1
6    1986-01-01    2016-09-12 2020_dataset           1
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit isotype is_antigen_specific antigen        MFI MFI_normalised  unit
1     1     IgE               FALSE   Total 1110.21154       2.493425 UG/ML
2     1     IgE               FALSE   Total 2708.91616       2.493425 IU/ML
3     1     IgG                TRUE      PT   68.56614       3.736992 IU/ML
4     1     IgG                TRUE     PRN  332.12718       2.602350 IU/ML
5     1     IgG                TRUE     FHA 1887.12263      34.050956 IU/ML
6     1     IgE                TRUE     ACT    0.10000       1.000000 IU/ML
  lower_limit_of_detection
1                 2.096133
2                29.170000
3                 0.530000
4                 6.205949
5                 4.679535
6                 2.816431
```

Q6. How many different Ab isotypes are there?

```r
unique(ab_data$isotype)
```

```
[1] "IgE"  "IgG"  "IgG1" "IgG2" "IgG3" "IgG4"
```

Q7. How many different Antigens are there in the dataset?

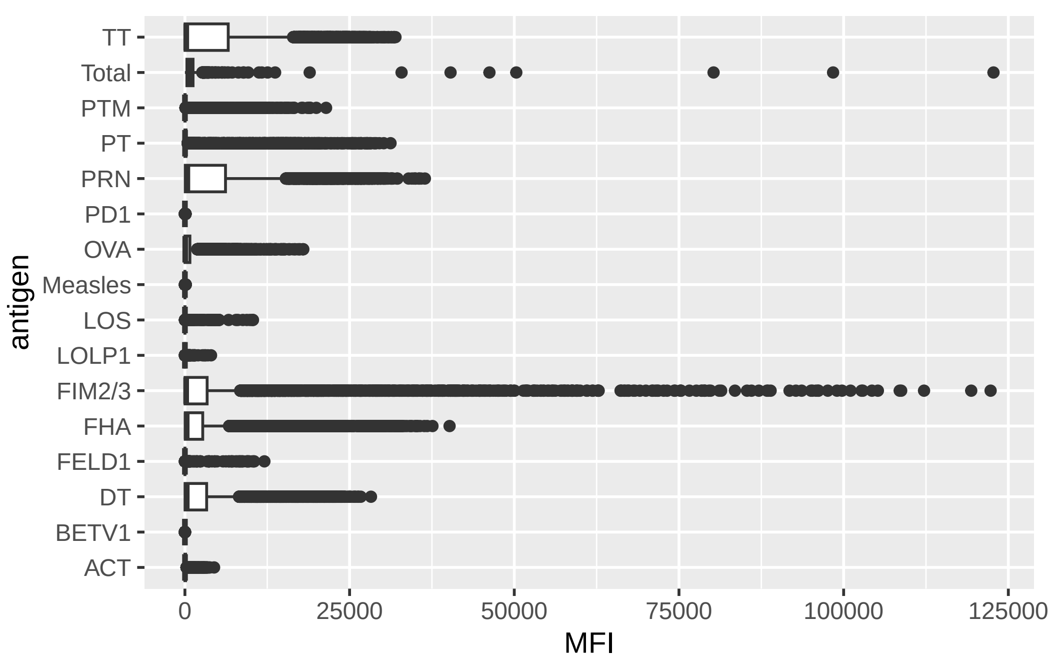
```r
unique(ab_data$antigen)
```

```
 [1] "Total"   "PT"      "PRN"     "FHA"     "ACT"     "LOS"     "FELD1"
 [8] "BETV1"   "LOLP1"   "Measles" "PTM"     "FIM2/3"  "TT"      "DT"
[15] "OVA"     "PD1"
```

Q8. Let's plot antigen MFI levels across the whole dataset

```r
ggplot(ab_data) +
  aes(MFI, antigen) +
  geom_boxplot()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_boxplot()`).
```



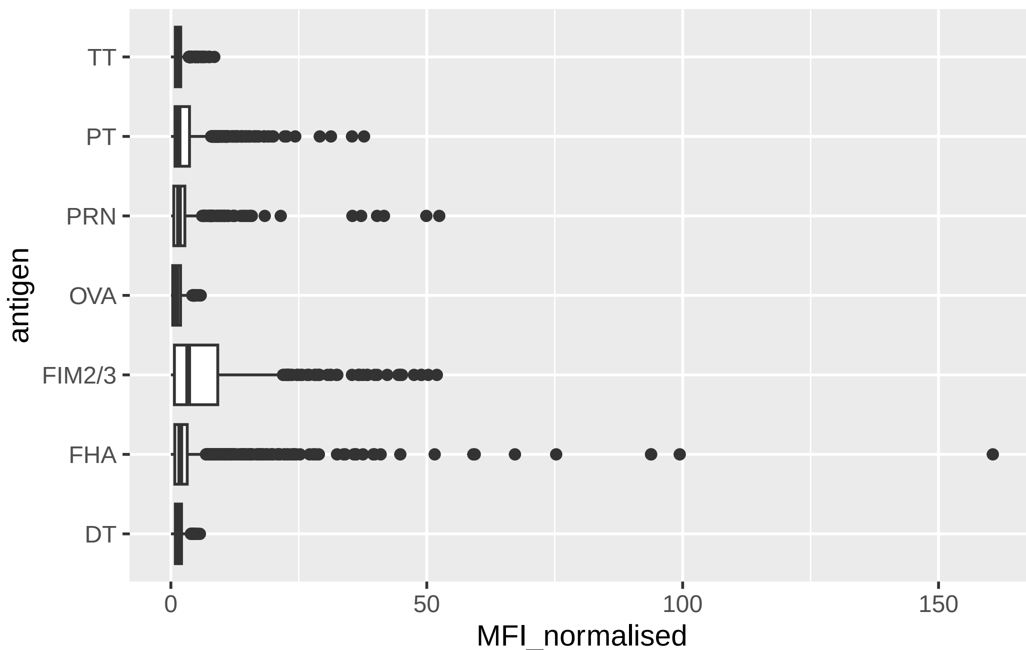
Focus in IgG

IgF is crucial for long-term immunity and responding to bacterial and viral infections

```
igg <- ab_data |>
  filter(isotype=="IgG")
```

Plot of antigen levels again but for IgG only

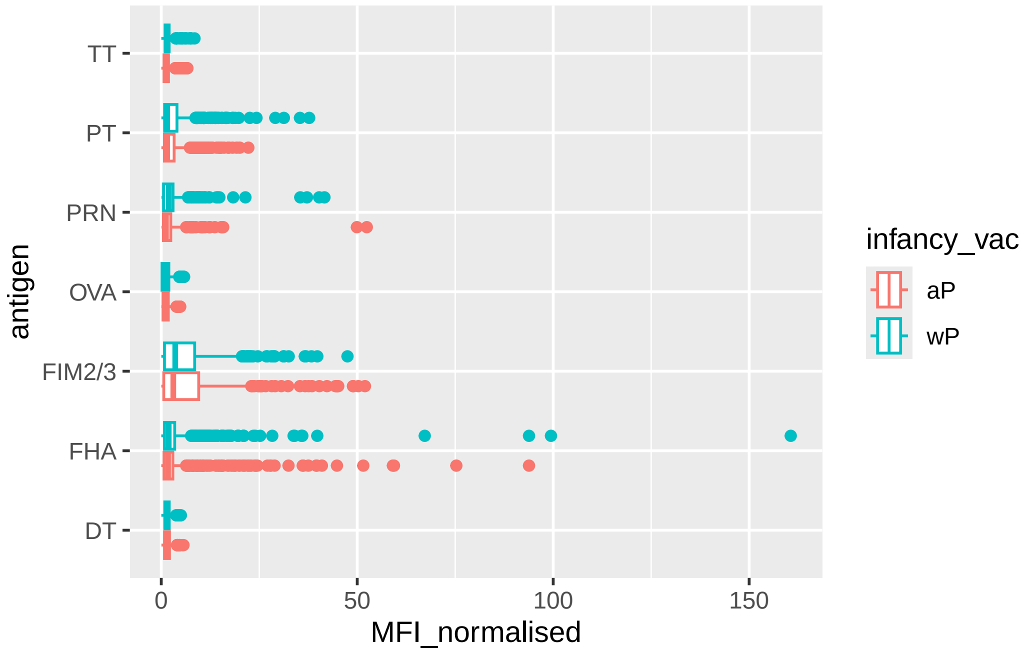
```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



Differences between aP and wP?

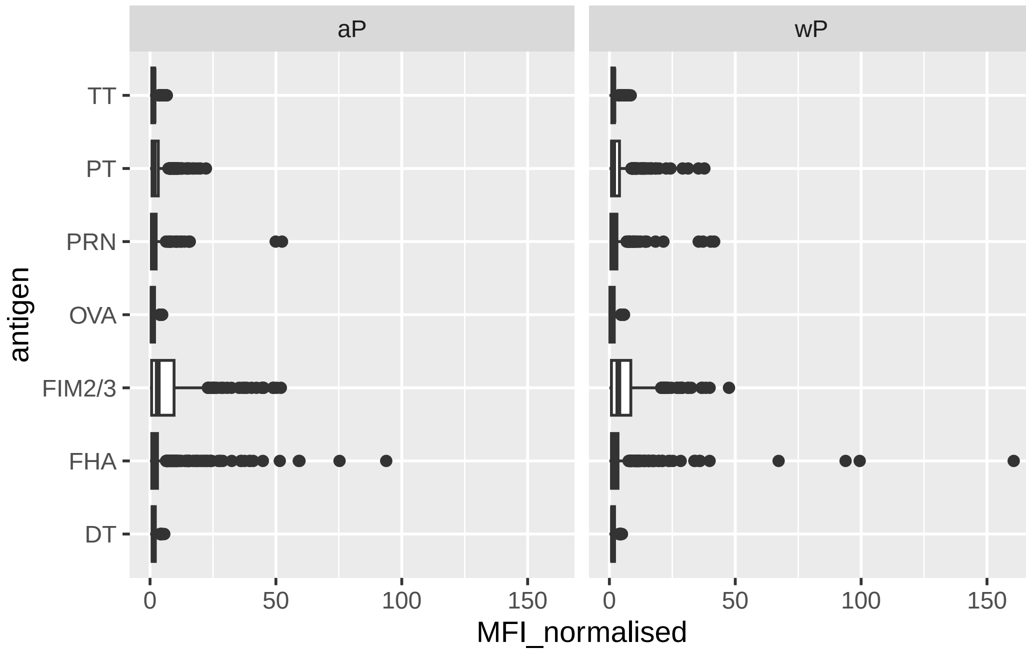
We can color up by the `infancy_vac` values of "wP" or "aP"

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



We could "facet" by the "aP" vs "wP" column

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



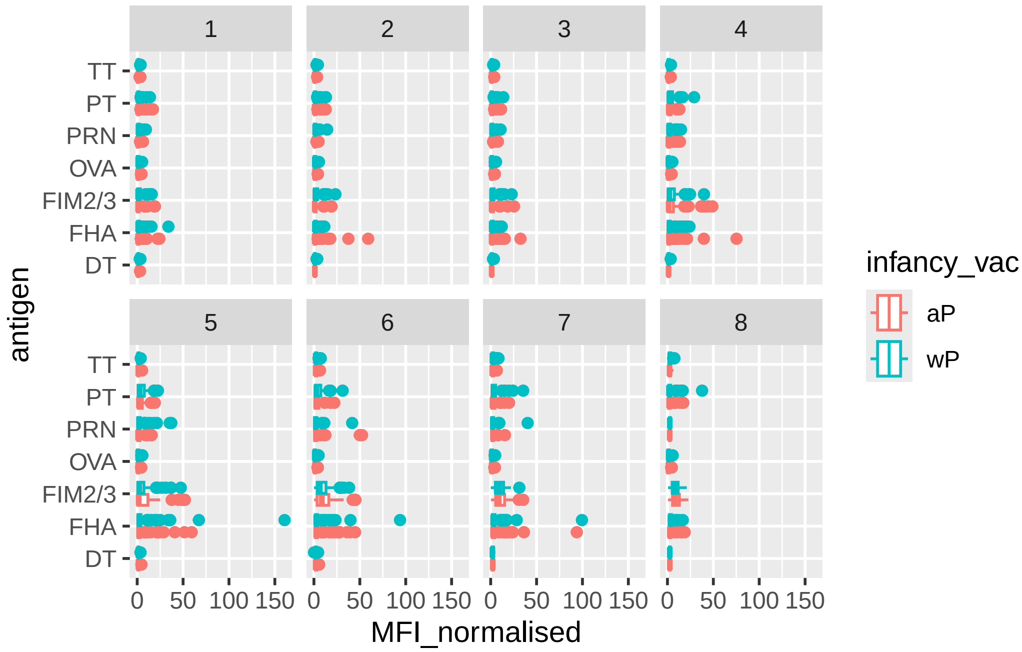
Time course analysis

We can use `visit` as a proxy for time here and facet our plots by this value 1 to 8

```
table(ab_data$visit)
```

```
   1    2    3    4    5    6    7    8    9   10   11   12
8280 8280 8420 8420 8420 8100 7700 2670  770  686  105  105
```

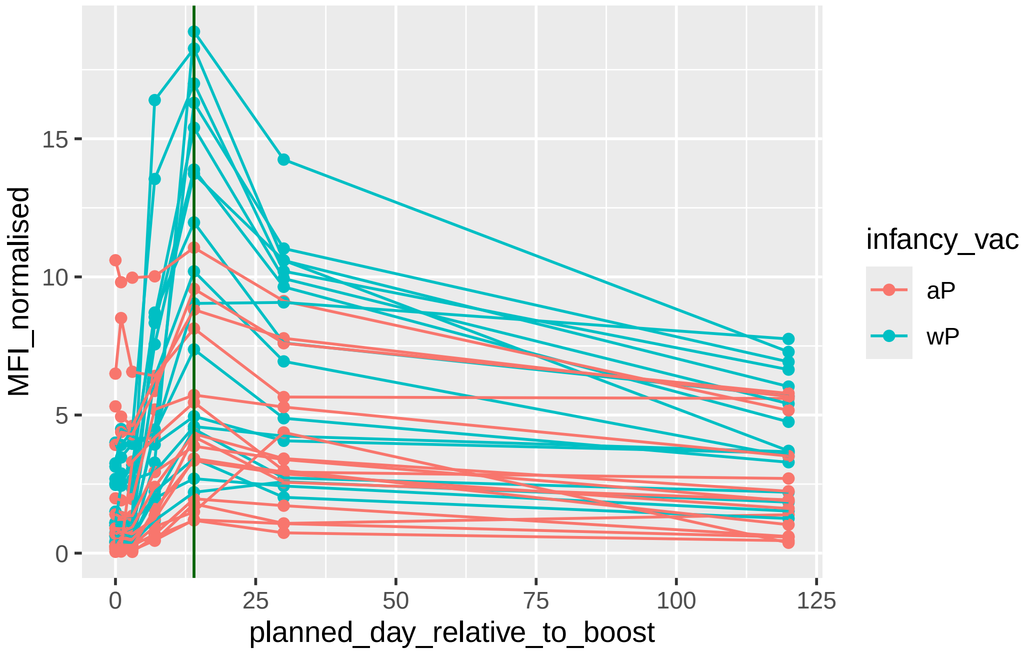
```
igg |>
  filter(visit %in% 1:8) |>
  ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit, nrow=2)
```



Time course of PT (Virulence Factor: Pertussis Toxin)

```
pt <- igg |>
  filter(antigen == "PT") |>
  filter(dataset == "2021_dataset")
```

```
ggplot(pt) +
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col=infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 14, col="darkgreen")
```



System setup

```r
sessionInfo()
```

```
R version 4.5.1 (2025-06-13)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.7.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base
```

```
other attached packages:
[1] dplyr_1.1.4    jsonlite_2.0.0 ggplot2_4.0.1

loaded via a namespace (and not attached):
 [1] vctrs_0.6.5        cli_3.6.5          knitr_1.50         rlang_1.1.6
 [5] xfun_0.54          generics_0.1.4     S7_0.2.1           labeling_0.4.3
 [9] glue_1.8.0         htmltools_0.5.8.1  scales_1.4.0       rmarkdown_2.30
[13] grid_4.5.1         evaluate_1.0.5     tibble_3.3.0       fastmap_1.2.0
[17] yaml_2.3.10        lifecycle_1.0.4    compiler_4.5.1     RColorBrewer_1.1-3
[21] pkgconfig_2.0.3    rstudioapi_0.17.1  farver_2.1.2       digest_0.6.38
[25] R6_2.6.1           tidyselect_1.2.1   pillar_1.11.1      magrittr_2.0.4
[29] withr_3.0.2        tools_4.5.1        gtable_0.3.6
```